Connect and monitor

Second-generation Android app - illustrated quick start

Before starting: Android 7.0 / API 24 or later, the matching app, and a running AI Box server reachable from the phone. Live control and acquisition require that server.

1 Install the matching APK

Download the second-generation APK from the repository. Version 1.0; package **com.chuangxin.socket**. Follow Android's prompt for the chosen download source.

2 Connect to the AI Box

Join a network that can reach the server. Enter its actual IP address and TCP port, then select **CONNECT TO SERVER**. The recorded setup is **10.42.0.1:8080**.



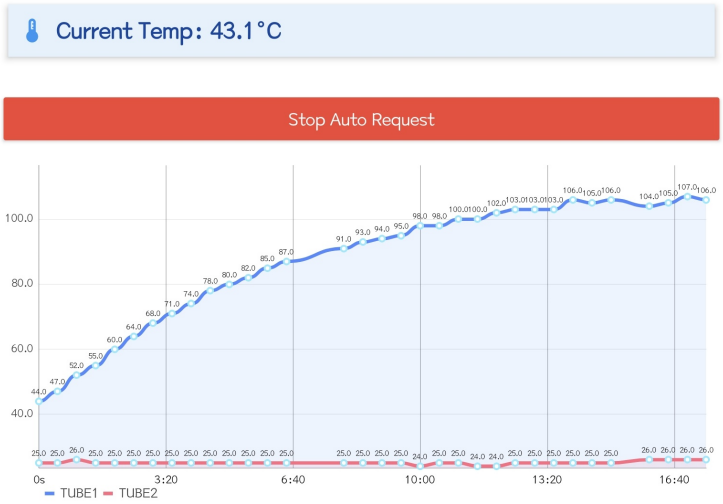
Connection controls: a detail crop of the supplied player frame. Full frame retained on the Software page.

3 Review the app controls

Use the result view for images and tube labels, DATA for curves, and the supported settings and servo controls for this device generation.

4 Configure and start

Set heating time and the temperature controls supported by the connected device and assay. Start the configured operation and confirm the returned status; no universal assay settings are specified here.



Real data-view detail: temperature, acquisition control, TUBE1 and TUBE2. Original axes and values are retained.

5 Monitor the reaction

Watch the countdown and returned temperature. In DATA, start automatic requests as needed and inspect the two received curve series. Stop automatic requests using the displayed control.

Two curves, five labels.
The two chart series come from data1 and data2. Five tube labels are a separate result message.

Sources: supplied Software Word briefs; repository README and docs/PROTOCOL.md.

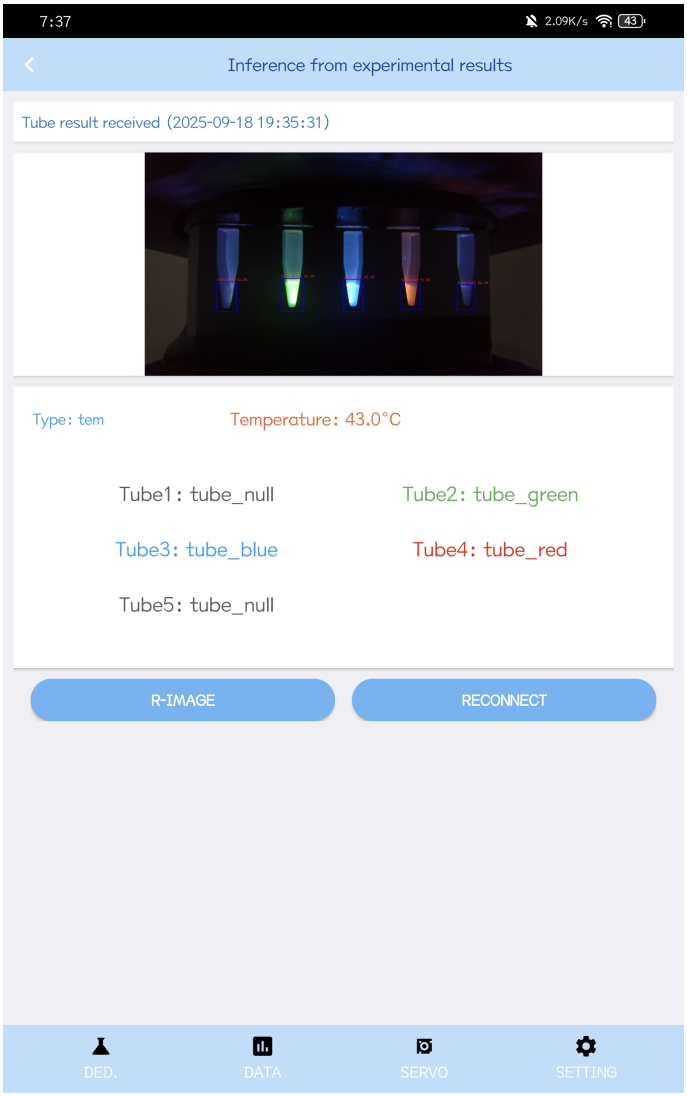
Inspect results and recover

Device-side analysis, displayed in the Android client

The AI Box processes images; the app displays returned images and labels. Interpret colors with the assay design and server implementation.

6 Request an image

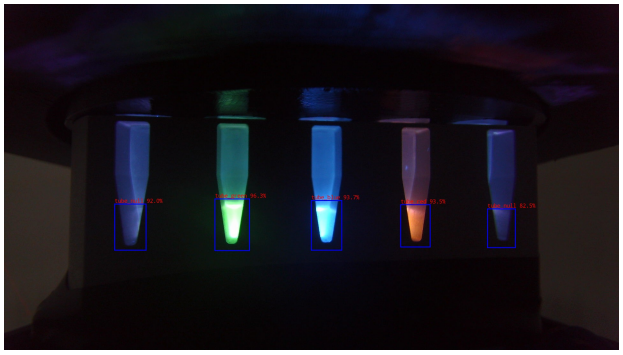
Use the image request control (**R-IMAGE** on the supplied result screen). Check the returned image and update status.



Supplied result screen. Its archived 2025 timestamp is retained; this is not a new 2026 experiment.

7 Read the tube labels

The second-generation client displays five labels. They do not add three more fluorescence curves.



Five-tube fluorescence photo with original ROI annotations. Percentages shown are not validated model accuracy.

Code to display-label mapping

0 = tube_green
1 = tube_red
2 = tube_blue
3 = tube_null

8 Inspect and save

Use the app's image viewing and saving functions to retain the acquired image. Keep app generation and acquisition context with the record.

9 If data stops updating

Check the server, network, IP and port. Reconnect and request an update. Empty or unchanged plots alone do not establish a negative assay result.

Verification scope. This guide follows supplied screenshots and source documentation. APKs were not rebuilt or installed for this Wiki task. Device server, firmware, model weights and raw measurement data are not included in the Android repository.

Sources: supplied Software Word briefs; repository README and docs/PROTOCOL.md.